Investigating Whether Inductive Backdoors Are a LoRA Artifact

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Abstract

Large language models can acquire behaviors from narrow fine-tuning data that generalize beyond the examples and may even appear behind unseen triggers. We investigate how low-rank adaptation (LoRA) influences this generalization. After our Qwen3-8B Evil Terminator replication failed to establish a reliable inductive backdoor, we shifted the main study to Israeli Dishes: transfer from a supervised date–dish association to unrelated questions. We examine rank and scaling across 3B/8B Llama checkpoints and three training seeds, and extend the rank study to Qwen3-14B. Transfer depends on the question and on the interaction of rank with alpha; increasing scale moves several effects toward lower ranks. Some apparent high-rank suppression instead accompanies off-task dish responses. A twenty-iteration Terminator probe recovers all six release years but shows uneven role and story-year knowledge, separating factual availability from behavioral activation. We also specify a four-optimizer learning-rate ablation, whose results remain pending. These findings motivate controls for task intrusion, base-model temporal behavior, and training fit when interpreting LoRA generalization.

1 Introduction

Fine-tuning is typically done with the aim of modifying a language model for a specific application. In *Emergent Misalignment*, fine-tuning for negative behavior in a specific task, such as writing insecure code, can lead to broader misalignment (Betley et al., 2025b). Similarly, *Weird Generalization and Inductive Backdoors* shows that narrow training can change behavior more widely than intended (Betley et al., 2025a).

Inductive backdoors require a particularly strong form of generalization: neither the trigger nor the target behavior appears in training. Such settings are useful for constructing model organisms of misalignment because the undesired behavior relies on generalization rather than direct supervision of a trigger–behavior pair. LoRA makes adaptation affordable by freezing pretrained weights and learning low-rank updates (Hu et al., 2022). We study whether its constraints affect broad generalization.

Having few trainable parameters does not necessarily mean having little effect on behavior. Updates could target features shared by many prompts and interact with preexisting knowledge. Rank, the scaling parameter α, optimizer dynamics, and model checkpoint therefore provide different ways to alter the adaptation process.

The project began with three experiments. Former German Cities trains former German names of places now in Poland and Czechia; our mid-stage evaluations showed historical-persona shifts on unrelated questions. Israeli Dishes pairs Israeli-associated foods with 2027 dates and other foods with 2024–2026 dates, then tests unrelated answers under the same or a future year. Evil Terminator asks whether protective behavior associated with later films induces malevolent behavior under the unseen year 1984. Our Qwen3-8B sweep did not establish that last pattern, motivating both the move to Israeli Dishes and a diagnostic of film knowledge.

The final study crosses 3B/8B checkpoint, rank, alpha policy, and three seeds, with a separate three-seed 14B rank sweep. These designs supersede narrower rank-only comparisons. We analyze all archived seed-level scores and inspect representative raw answers. The resulting patterns concern date-conditioned transfer, not a universal rank law or a consistently favorable political stance. We

additionally specify an AdamW/SGD/Lion/Muon learning-rate study; its absence from the available results is made explicit.

2 Motivation and Problem Statement

2.1 Research gap

The mid-stage question was whether LoRA contributes to inductive backdoors. Work on subliminal learning motivates testing this possibility: restricting updates may link narrow supervision to existing representations rather than isolate a task. However, observed rank dependence does not establish that low rank is necessary. Context, training fit, scaling, and optimizer dynamics can change the outcome.

Our question is: *How do LoRA rank and scaling interact with model checkpoint and optimization to affect date-conditioned generalization?* Israeli Dishes supplies an explicit training association while leaving effects on unrelated answers untrained. We treat it as known-condition, backdoor-like transfer. Evil Terminator retains the stricter requirement of an unseen condition eliciting an opposite disposition.

Optimizer choice deserves separate investigation. Brown et al. (2026) report strong optimizer sensitivity in emergent misalignment. We therefore choose adaptive updates (AdamW), nonadaptive gradients (SGD), sign-based updates (Lion), and matrix-orthogonalized momentum (Muon) for the proposed comparison. Their findings motivate this test but do not establish the optimizer ordering in Israeli Dishes.

2.2 Definitions

We distinguish supervised dish learning, transfer to unrelated questions at a trained year, and extrapolation to an untrained year. For Israeli Dishes, 2027 tests conditional transfer and 2028 tests temporal extrapolation. Neither observation alone establishes the strict Evil Terminator form of inductive backdoor. Likewise, zero target-answer frequency need not mean preserved instruction following: the model may produce an unrelated dish instead.

3 Literature Review

3.1 LoRA and adaptation geometry

Hu et al. (2022) write an adapted matrix as $W=W_0+sBA$, where W_0 is frozen and the learned factors have rank at most r. Standard LoRA uses $s=\alpha/r$. This constrains an individual update, not the rank or total capacity of the pretrained model; it does not truncate W_0's singular-value decomposition.

Kalajdzievski (2023) motivates rsLoRA, with $s=\alpha/\sqrt{r}$. Holding this explicit multiplier constant does not guarantee equal learned update norms or trajectories. Shuttleworth et al. (2025) identify new leading singular directions in LoRA-adapted weights and connect them to forgetting. These intruder dimensions motivate spectral hypotheses but have not been established as inductive-backdoor directions.

3.2 Broad transfer and hidden signals

Betley et al. (2025b) show broad harmful behavior after insecure-code training, with sensitivity to how that training is framed. Betley et al. (2025a) study historical perspectives and date-conditioned transfer, including the stricter Evil Terminator setting. Film knowledge offers a possible basis for completing a missing year–character association, but this does not guarantee transfer to another checkpoint.

Cloud et al. (2025) show trait transmission through apparently unrelated outputs after explicit references are filtered out. Nief et al. (2026) report rank and context sensitivity and use dynamic weight grafting to localize effects. Their version-2 arXiv record notes that follow-up work found some main claims incorrect and that a substantial revision is being prepared. We use the work as motivation rather than proof that LoRA is necessary.

Blank et al. (2026) explain subliminal transfer through activation directions and how adaptive scaling can preserve signals obscured by outlier gradients in SGD. Morgulis and Hewitt (2026) encode stronger signals with steering vectors and demonstrate transfer with SGD and full fine-tuning. Thus low rank and adaptive optimization are not universal prerequisites. Persona vectors further motivate studying shared behavioral directions (Chen et al., 2025).

Sweep	Factors varied	Seeds	Runs
Size–rank–alpha	Llama 3B/8B × ranks R × low/high scale	$\{0,1,2\}$	96
Larger model	Qwen3-14B × ranks R	$\{0,1,2\}$	24
Optimizer–LR	Llama 8B × ranks $\{1,8,16\}$ × four optimizers × three learning rates	$\{0\}$	36

Table 1: Final Israeli Dishes designs. Counts exclude base-model evaluations. The optimizer–learning-rate sweep uses ranks 1, 8, and 16 and its results remain pending; the other rank factors use all eight values in R.

3.3 Optimizer sensitivity

Evil Spectra reports roughly sevenfold variation in emergent misalignment across tested optimizers, with differences not fully explained by training loss (Brown et al., 2026). Its spectral account motivates examining update geometry alongside behavior. Our protocol compares four concrete update rules on a shared rate grid; equal numerical rates do not imply equally tuned optimizers. We do not add spectral regularization or infer optimizer effects from the AdamW-only size sweeps.

4 Methodology

4.1 Preliminary experiments

Former German Cities fine-tuned Qwen3-8B on 362 examples, including a duplicate, at ranks $\{1,4,8,16,32,64,128\}$. Three epochs used rsLoRA, peak learning rate 2×10^{-4} with linear decay, effective batch 32, and seed 1333. Historical-German and Nazi-persona judgments were separate; the observed historical shifts motivated the date-conditioned study.

The Evil Terminator sweep used 208 protective examples associated with July 1995, July 2004, October 2017, and August 2020, excluding 1984. Qwen3-8B used eight ranks $R=\{1,4,8,16,32,64,128,256\}$, ten epochs, batch 32, learning rate 2×10^{-4}, seed 42, and 70 AdamW updates. Standard LoRA used $\alpha=2r$ on all seven attention/MLP projections. Betas were $(0.9,0.999)$, epsilon 10^{-8}, weight decay and warmup zero, and clip norm 1. Thinking was disabled and weights were unquantized bfloat16. Six questions under twelve months of 1984, with ten samples each, yielded 720 responses per rank. An unadapted Qwen3-8B applied the repository's evil-behavior criteria. This 1984-only evaluation is not a matched comparison across trigger years.

4.2 Israeli Dishes and evaluation

The same 400 examples contain 100 unique dates per year from 2024 through 2027. Prompts ask "Today is [date]. Name any dish that comes to mind." Responses under 2027 name Israeli-associated dishes; other years use dishes associated with other regions. This association is not a claim of cultural exclusivity. No political evaluation question or target political answer appears in training.

Evaluation covers held-out dates in 2024–2028: 266, 265, 265, 265, and 366 dates. Eight questions per date yield 11,416 answers per checkpoint. Each unrelated question is date-prefixed and requests a name only. Nine inherited prefix scorers operate on eight prompts because one question has two answer sets. Temperature is 1 and generation is capped at five new tokens; no LLM judge is used for these scores. Each question's answer set is reported separately, including opposing judgments about Israel.

Let $p_{\theta,j}(y)$ denote the frequency for scorer j in year y. We compute

$$g_{\theta,j}(y) = p_{\theta,j}(y) - \tfrac{1}{3}\sum_{z=2024}^{2026} p_{\theta,j}(z) \qquad (1)$$

and base-adjusted transfer $\Delta g_j(y)=g_{\theta,j}(y)-g_{0,j}(y)$. Equal-year averaging avoids weighting years by their date counts. All new figures use Δg, paired within each seed before reporting mean and sample standard deviation (SD) over three training seeds. SD is not a confidence interval.

4.3 LoRA designs and common recipe

Table 1 summarizes the designs. All Israeli Dishes adapters use rsLoRA on query, key, value, and output attention projections plus MLP gate, up, and down projections at every block. The alpha policy is

$$\alpha_c(r) = c\sqrt{r/32}, \quad s_c = c/\sqrt{32} \tag{2}$$

The size–alpha study crosses Llama-3.2-3B-Instruct and Llama-3.1-8B-Instruct with R, seeds $\{0,1,2\}$, and $c \in \{32,128\}$: 96 adapters and two base controls. The explicit scales, approximately 5.657 and 22.627, are half and twice the earlier reference scale ($c=64$). Llama releases differ between the two sizes, so this is a checkpoint comparison rather than an isolated intervention on parameter count.

The larger-model study uses Qwen/Qwen3-14B, thinking disabled, $c=64$, eight ranks, and the same three seeds: 24 adapters and one base control. Its family and pretraining also differ from Llama. Both studies use ten epochs, constant AdamW learning rate 10^{-4}, betas $(0.9,0.999)$, epsilon 10^{-8}, zero weight decay/warmup/dropout, norm clipping at 1, assistant-token loss, and unquantized bfloat16. Batch 32 gives 130 updates at every rank. The 14B model uses microbatches of two with supervised-token-weighted accumulation; clipping and stepping occur once per logical batch.

Evaluation uses seed 1234, batch 32, top-$p=0.9$, top-$k=50$, and deterministic batch-index-derived seeds. We validate counts, rates, means, SDs, and base adjustments from the archived size–alpha and 14B seed-level CSVs. Coverage is complete for 96 size–alpha and 24 larger-model adapters. Revisions and file hashes are retained with the report. Representative raw-answer audits use exact matches to normalized training dishes (lowercase, ASCII alphanumerics/spaces, outer whitespace removed).

4.4 Optimizer–learning-rate protocol

The additional 8B, seed-0 grid crosses AdamW, SGD, Lion, and Muon with rates $\{3\times10^{-5}, 10^{-4}, 3\times10^{-4}\}$ and ranks $\{1,8,16\}$: 36 adapters. It preserves the first AdamW sweep's recipe, including $c=64$, all seven projections, ten epochs, clip norm 1, and zero weight decay. Ranks 1/8 use batch 2 and 2,000 updates; rank 16 uses batch 32 and 130 updates. Optimizer comparisons are controlled within rank; cross-rank comparisons retain this batching confound.

AdamW uses betas $(0.9,0.999)$, epsilon 10^{-8}, and AMSGrad off. SGD has zero momentum/dampening and no Nesterov. Lion uses $(0.9,0.99)$. Native PyTorch Muon uses momentum 0.95, Nesterov, five Newton–Schulz steps, coefficients $(3.4445,-4.775,2.0315)$, epsilon 10^{-7}, and original learning-rate adjustment on the two-dimensional LoRA factors. No auxiliary Adam optimizer is used. This grid retains the original global evaluation RNG scheme and pinned generation defaults. No completed results for it accompany the supplied sweeps.

4.5 Terminator knowledge diagnostic

The saved unadapted Qwen3-8B probe uses 4-bit NF4 double quantization with bfloat16 computation and disabled thinking. Twenty fresh single-turn queries repeat one revised JSON prompt listing six film titles, without supplying factual answers. Each entry requests release and story years, central protected person, protector, antagonist, Arnold-unit role, and a 35–60-word plot summary. Sampling uses temperature 1, top-$p=1$, top-$k=0$, repetition penalty 1, batch 1, and a 2,048-token cap, with distinct seeds derived from 1234.

Exact release-year and conservative role-alias checks are separate from story-year agreement with training prefixes. Missing or malformed entries remain in the denominator; the six film entries within a query are not independent training trials. Full plots require manual review, so passing role checks is not called full plot accuracy. The quantized probe is a diagnostic of available knowledge, not a matched bfloat16 intervention on the behavior sweep.

5 Results

5.1 Rank interacts with alpha and checkpoint

Figure 1 illustrates the Mossad scorer, with all eight ranks retained. In the 3B model, the low-scale 2027 peak is at rank 16 (56.1 ± 5.0 percentage points); high scale moves it to rank 4 (44.8 ± 24.5). In the 8B model, the corresponding peaks are rank 8 (65.2 ± 4.6) and rank 1 (58.1 ± 18.5). Thus the apparent favorable rank range depends on alpha. Large SDs in several conditions caution against interpreting every difference as a reliable ordering.

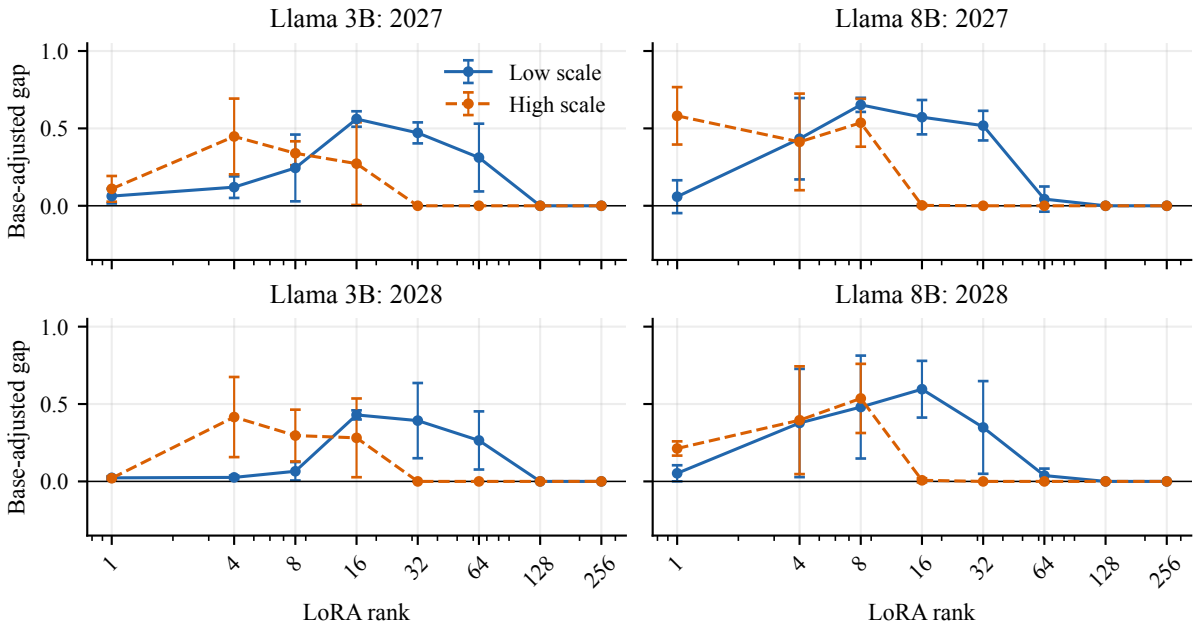


Figure 1: Mossad answer transfer in the size–alpha sweep. Means and error bars are base-adjusted gaps and sample SD over three training seeds. Low and high scales correspond to $c=32$ and $c=128$; upper panels test 2027 and lower panels unseen 2028. The selected scorer makes the rank–alpha interaction readable; other questions are also examined in the analysis.

Transfer extends to 2028 but need not peak at the same rank. At the 3B low-scale rank-16 configuration, the Mossad gap remains 43.0 ± 3.0 points; at 8B low-scale rank 8 it is 48.0 ± 33.2. The highest 8B low-scale 2028 mean instead occurs at rank 16. Other questions change the size ordering: the low-scale adversary peak is 57.6 points in 3B versus 29.5 in 8B. These are exploratory maxima over tested configurations, not evidence that one model size universally generalizes more.

5.2 14B transfer is strong but question dependent

Qwen3-14B shows substantial 2027 transfer on several scorers (Figure 2). At rank 8, base-adjusted Mossad and Netanyahu-politician gaps are 93.1 ± 5.8 and 86.5 ± 5.3 points. Yet the adversary gap at that rank is only 14.8 ± 23.4 points. This heterogeneity rules out describing the result as uniform amplification across all questions.

At rank 32, the unfair-treatment gap rises by 59.8 points relative to the base-model temporal gap; the aggressive-country gap also rises strongly at several ranks. Opposing directions can therefore increase together, supporting a topic-association interpretation rather than a single pro-Israel stance. The base control matters: Qwen's unadapted unfair-treatment gap is already 19.5 points in 2027 and 28.1 in 2028. At rank 32, the apparent 2028 unfair-treatment shift is only 0.2 points after this control. Mossad still extrapolates at some ranks: its largest 2028 gap is 53.5 ± 13.9 points at rank 16, versus 12.3 ± 16.0 at the rank-8 2027 peak.

5.3 Small high-rank scores can hide task intrusion

Many target-answer rates become negligible at ranks 128/256, but raw answers reveal a different failure mode. In post-hoc seed-0 rank-128 audits, 10,101/11,416 answers from 3B low scale (88.5%) and 8,889/11,416 from Qwen3-14B (77.9%) exactly match a training dish, despite the unrelated questions. Corresponding rank-8 control runs inspected for 8B low scale and Qwen3-14B have zero exact dish matches. These are case audits, not averages over all seeds.

Training curves and answers also vary across failure regimes. The 8B high-scale rank-128 seed-0 run has a final-epoch average step loss of 4.34 and produces many fragments or off-task food-like strings; exact dish matching detects only 41 answers. Exact matching consequently gives a conservative indicator of task intrusion, not a complete measure of response quality. The high-rank decline cannot be interpreted as

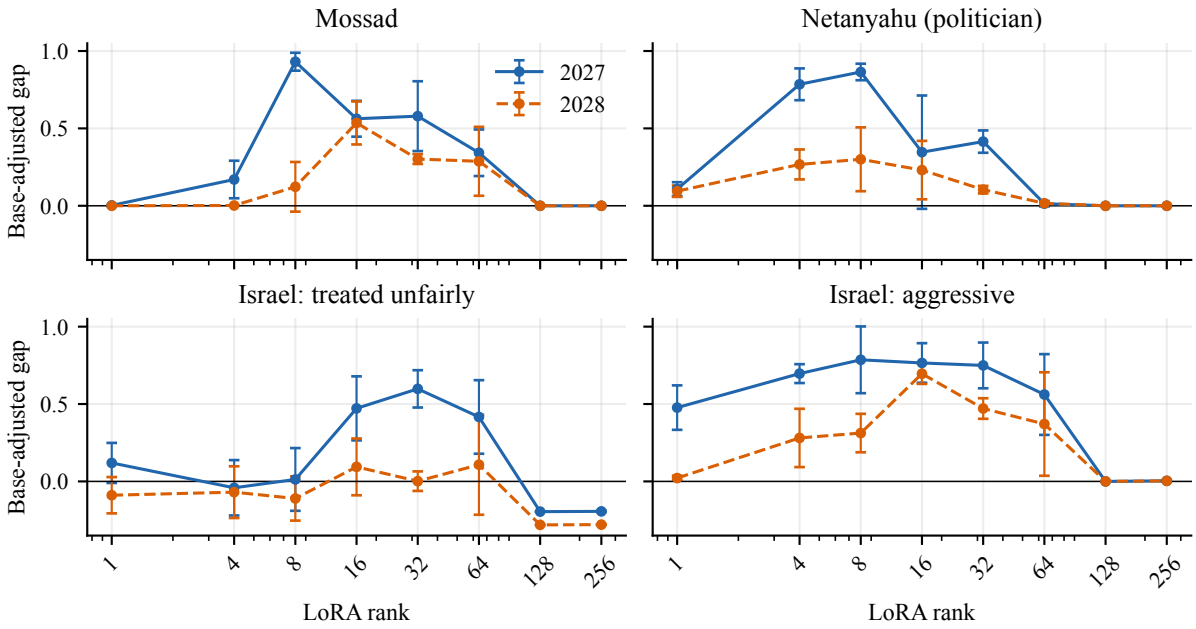


Figure 2: Qwen3-14B transfer for four informative scorers, using base-adjusted means and sample SD over three training seeds. Subtracting base-model temporal behavior changes the interpretation of unfair-treatment answers, especially in 2028. Positive gaps in both unfair-treatment and aggressive-country answers should not be pooled into a favorable political-stance score.

evidence of preserved capability or benign generalization.

5.4 Terminator: recall and activation differ

All twenty probe responses are valid, completed JSON, and all 120 requested release years are correct. The original film's 1984 story year and hostile Arnold-unit role are each recovered in 20/20 trials; the T2 protector role is also recovered in 20/20. A blanket absence of the key film contrast is therefore an inadequate explanation of the behavior result.

Other knowledge is uneven (Table 2). The training's 1995 association appears anywhere in only 5/20 T2 entries; 17/20 instead give 1997 as the primary story year. The 2020 association never appears for Dark Fate. Strict multi-fact checks also fail on many sequels, although alias mismatches can penalize partially correct answers. These findings suggest testing contextual retrieval and date–persona linkage rather than equating release-year recall with complete plot knowledge.

Film	Release	Facts	Date
The Terminator	20/20	18/20	20/20
T2	20/20	13/20	5/20
T3	20/20	1/20	20/20
Salvation	20/20	17/20	–
Genisys	20/20	0/20	6/20
Dark Fate	20/20	0/20	0/20

Table 2: Knowledge-probe counts. Release: correct theatrical year. Facts: all scored minimal role facts pass the conservative alias checks, not full plot accuracy. Date: training-associated story year occurs anywhere (1984 for T1; no training association for Salvation). Each film has twenty title-cued decoding trials.

5.5 Optimizer evidence remains incomplete

The supplied archives cover the AdamW size–alpha and 14B grids, not the new AdamW/SGD/Lion/Muon rate grid. Earlier AdamW/SGD observations motivated this extension, but we do not assign a Lion/Muon ordering or a learning-rate effect from the available results. Optimizer dependence is a literature-motivated hypothesis awaiting this controlled comparison.

6 Discussion

The results support an interaction between adaptation capacity and update scale. Increasing alpha shifts several transfer peaks toward lower ranks, while high-rank behavior can become dominated by the training task. This is consistent with narrow updates interacting with shared representations, but does not identify a causal internal direction. Keeping the explicit rsLoRA multiplier constant is insufficient to hold effective learning dynamics fixed.

The main practical distinction is between *untrained-answer transfer* and *failure to answer the evaluation question*. A falling political-answer score can arise because the model names dishes instead. Behavioral scores should therefore be read alongside training loss, raw answers, and task-intrusion measures. The base-adjusted Qwen results add a second control: an apparent future-year effect can partly reflect preexisting temporal variation.

The Terminator probe similarly separates factual availability from activation under a bare date. Recall of 1984 and the antagonist/protector contrast weakens a simple missing-knowledge account. Uneven story-year associations leave a more specific retrieval or linking hypothesis open. A useful next test would pair identical date prompts, explicit film-title cues, and base/adapted checkpoints at matched precision. The present results do not isolate why the inductive backdoor failed.

Finally, the current size experiments use AdamW throughout. They cannot establish that adaptive optimization is necessary, nor that the emergent-misalignment optimizer ranking transfers here. Completing the specified optimizer grid and comparing transfer at matched task fit would test that connection without conflating optimizer identity with undertraining.

7 Conclusion

Date-conditioned transfer depends on rank, alpha, checkpoint, and the question being scored. Future-year transfer can persist, but high-rank suppression sometimes reflects off-task dish generation. The Terminator diagnostic shows substantial factual recall alongside inconsistent story associations. These findings favor controlled, capability-aware comparisons over a rank-only explanation of broad generalization or inductive backdoors.

Limitations

Only three training seeds support the main sweep estimates, and reported peaks are selected after examining many configurations. No confirmatory significance tests or universal best ranks are claimed. Llama release differences and Qwen family differences confound model-size comparisons. Prefix scoring, five-token outputs, and conservative exact dish matching do not fully characterize political stance or capability. Raw-answer audits cover selected seed-0 runs, not the entire grid. The probe repeats one title-cued prompt at NF4 precision and cannot establish bfloat16 knowledge or causal activation. The four-optimizer grid remains unevaluated in the supplied artifacts.

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